

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1933

Windom 138 kV
September 26, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1933. The Interconnection Customer proposes a 128.7 MW solar photovoltaic facility interconnecting at the Windom 138 kV substation in the Maryland area. The request was received on November 16, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 702 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,440,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
VERMONT YANKEE to KLEEN 138 KV line	INTERNATIONAL PAPER VALLIANT OK SUBSTATION to KEARNEYVILLE 345 KV	119.9	274	7.3
VERMONT YANKEE to RIDGELEY 138 KV	INTERNATIONAL PAPER VALLIANT OK to KEARNEYVILLE SUBSTATION 345 KV	118.4	280	7.3
BEDFORD ROAD SUBSTATION to KLEEN 138 KV	INTERNATIONAL PAPER VALLIANT OK to KEARNEYVILLE 345 KV	109.5	266	7.3
CLARK COUNTY 345-138 kV Transformer	ARTESIA COUNTRY CLUB to CLARK COUNTY 345 KV	103.6	845	5.7
THIELE KAOLIN REEDY CRK 500-345 kV Auto	KEARNEYVILLE to INTERNATIONAL PAPER VALLIANT OK 345 KV	101.9	802	14.4
ARTESIA COUNTRY CLUB 345/1kV TX	ARTESIA COUNTRY CLUB to AIR FORCE BASE 345 KV line	101.6	901	5.7
ARTESIA COUNTRY CLUB 138/1kV Auto	AIR FORCE BASE to ARTESIA COUNTRY CLUB 345 KV	101.6	901	5.7
E COOL SPRINGS to HACIENDA LAS CAROLINAS 138 KV L	SPINDLE to ALLIGATOR 500 KV	100.8	216	8.2
RONCO to FIRSTENERGY FORT MARTIN POWER STA 345 KV	AUBURN RD to PRIDES CORNER 345 KV	100.6	1,764	5.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
VERMONT YANKEE to KLEEN 138 KV line	Rebuild line with higher-capacity conductor	\$3,374,000	20% (\$684,000)
VERMONT YANKEE to RIDGELEY 138 KV	Rebuild line with higher-capacity conductor	\$5,670,000	20% (\$1,150,000)
BEDFORD ROAD to KLEEN 138 KV	Reconductor line	\$453,000	20% (\$92,000)
CLARK COUNTY SUBSTATION 345 KV/138 KV Auto	Replace transformer with higher-rated unit	\$21,906,000	2% (\$538,000)
THIELE KAOLIN REEDY CRK 500-345 KV Auto	Replace transformer with higher-rated unit	\$30,755,000	4% (\$1,100,000)
ARTESIA COUNTRY CLUB 345/1KV TX	Replace transformer with higher-rated unit	\$14,215,000	3% (\$379,000)
ARTESIA COUNTRY CLUB 138/1 KV TX	Replace transformer with higher-rated unit	\$4,865,000	3% (\$130,000)
E COOL SPRINGS to HACIENDA LAS CAROLINAS 138 KV L	Reconductor line	\$3,760,000	8% (\$313,000)
FIRSTENERGY FT MARTIN POWER STA to RONCO SUBSTATION 345 KV	Reconductor line	\$10,103,000	0% (\$54,000)
	Total allocated to project		\$4,440,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.